Student Number

PEM

2020

Year 11 End of Year Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time - 2 hours
- Write using black pen
- NESA approved calculators may be used
- A NESA reference sheet is provided
- In Questions 13 – 30, show relevant mathematical reasoning and/or calculations

Total Marks – **80**

Section I Questions 1 – 12 **12 marks**
Allow about 20 minutes for this section.

Section II Questions 13 – 30 **68 marks**
Allow about 1 hour and 40 minutes for this section.

Section I (12 marks)

Write your answers on the multiple choice answer sheet.

Allow about 20 minutes for this section.

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1 The chance of drawing a red card from a pack of 52 playing cards is

(A) likely
(B) impossible
(C) even
(D) certain

2 A score is added to the set of scores: 15,18,20,22,25,25. If the new mean is 21, the score that was added is

(A) 22
(B) 21
(C) 25
(D) 20

3 A microbe is about 1/10th the length of a typical human cell. The average length of a human cell is 100 micrometres and there are 1 million micrometres in a metre. The average length of a microbe in metres would be

(A) 1×10^{7} m
(B) 1×10^{-6} m
(C) 1×10^{-5} m
(D) 10 m

4 The formula $4m = 5n + 2p^2$ with p as the subject would be

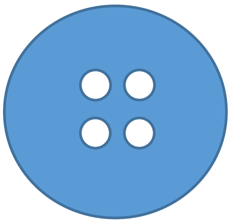
(A) $\pm\sqrt{\dfrac{5n-4m}{2}}$

(B) $\sqrt{\dfrac{4m-5n}{2}}$

(C) $\pm\dfrac{\sqrt{4m-5n}}{2}$

(D) $\pm\sqrt{\dfrac{4m-5n}{2}}$

5 A manufacturer of buttons needs to work out the shaded area shown below



If the diameter of the larger circle is 20mm and the four smaller circles have diameter 2mm each, the area of the face of the button is

(A) 99π mm^2
(B) 96π mm^2
(C) 384π mm^2
(D) 396π mm^2

6 Jin rolls a die 10 times and her scores are:

$$3 \quad 4 \quad 2 \quad 5 \quad 6 \quad 3 \quad 1 \quad 4 \quad 5 \quad 2$$

Su rolls the same die 10 times and her scores are

$$5 \quad 2 \quad 2 \quad 1 \quad 6 \quad 4 \quad 6 \quad 3 \quad 2 \quad 6\,.$$

To win the game you must have the greatest mean or median or both.
Which statement is true

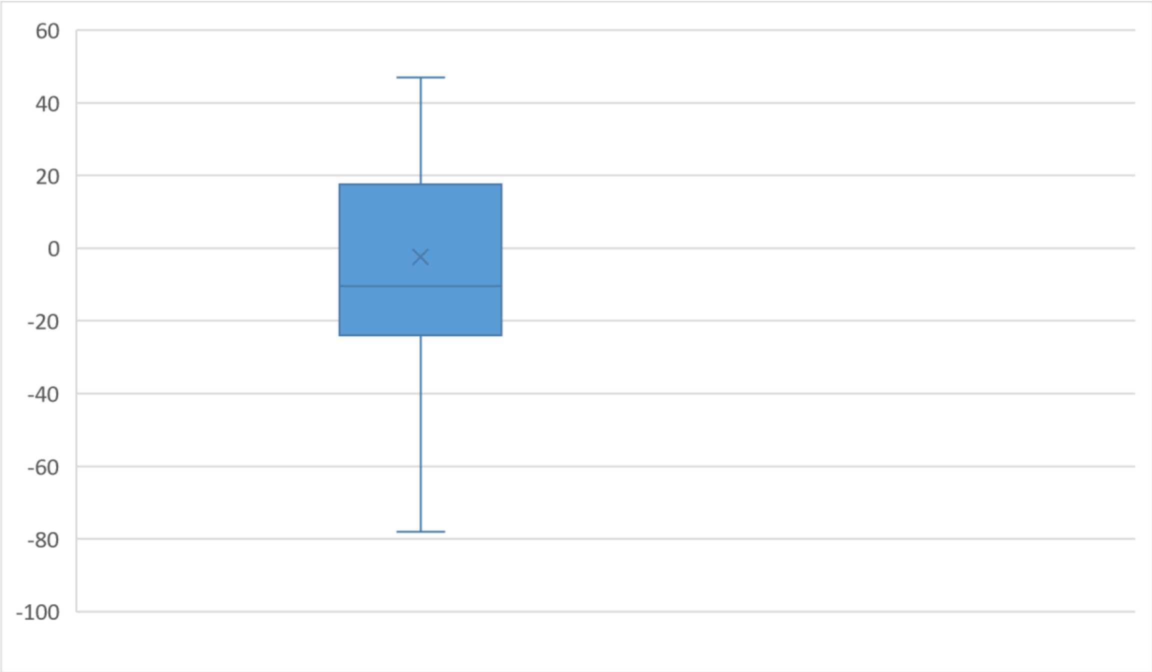
(A) Jin wins because her mean is greater than Su's
(B) Su wins because her mean is greater than Jin's
(C) Su wins because her median is greater than Jin's
(D) Su wins because both her mean and median are greater than Jin's

7 Joe arrives at his office in Sydney at 9:15am on Monday morning. He checks his international clock on his phone and it says that the time is 10:15pm on Sunday in London.

If he wants to phone his client at 9am on Monday, London time, what time should he call from Sydney?

(A) 8am Tuesday
(B) 10am Tuesday
(C) 10pm Monday
(D) 8pm Monday

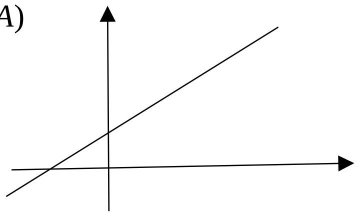
8 What is the median of the data set represented by the box plot?



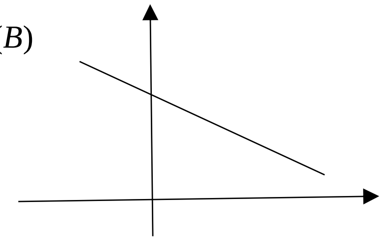
(A) $\quad 19$
(B) $\quad -21$
(C) $\quad -10$
(D) $\quad 0$

9 Identify the linear graph with a negative gradient.

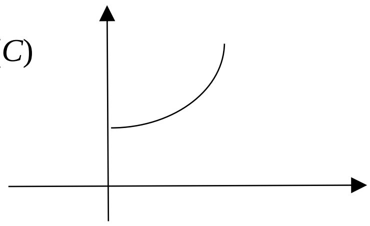
(A)



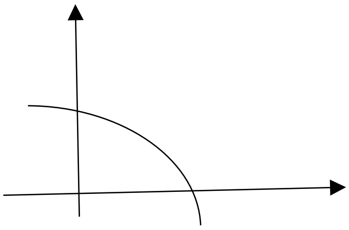
(B)



(C)



(D)

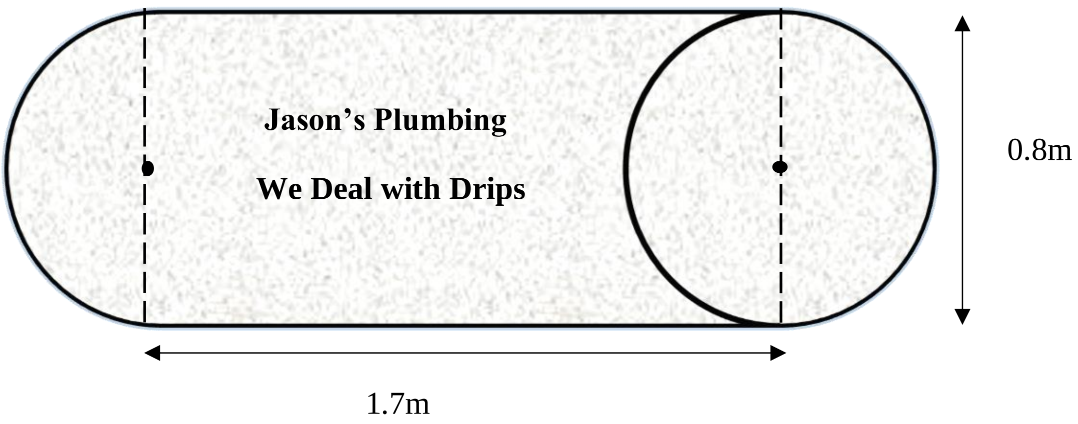


10 After completing his apprenticeship as a plumber Jason decides to start his own business. He purchases a van for \$35 000. The van depreciates at a rate of 25% p.a.

Find the salvage value of the van after 4 years when he wants to replace it.

(A) \$8750
(B) \$26250
(C) \$11074
(D) \$0

11 The 2D design below is the logo that will appear on the side of Jason's van.



Calculate the area of Jason's design for the side of the van.

(A) 1.36 m^2
(B) 1.86 m^2
(C) 4.27 m^2
(D) 3.37 m^2

12 Saria works at a café. To help with the ordering she decides to do a tally on one of her average days at the café.

Type of Milk	Number of Orders
Full cream milk	45
Skim milk	30
Soy milk	12
Almond milk	3

For this particular day, which one of following statements is **false**:

(A) Half the orders are for full cream milk.
(B) The probability that a customer orders almond milk is less than 3%.
(C) A customer is 4 times more likely to order soy milk than almond milk.
(D) Full cream is the favourable option.

END OF SECTION I

Student Number

Section II (68 marks)

Attempt Questions 13 - 30

Allow about 1 hour and 40 minutes for this section.

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided on pages **25** and **26**. If you use this space, clearly indicate which question you are answering.

Write your Student Number at the top of this page.

Please turn over

Question 13. (2 marks)

Evaluate the equation $c = \sqrt{a^2 + b^2}$ by substituting $a = 6\ and\ b = 8$. **2**

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Question 14. (3 marks)

A motorbike is for sale for $11 999. Julie buys the bike on the following terms:

Deposit	25%
Monthly repayments for 3 years	$405

(a) How much does Julie pay in total for the bike? **2**

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(b) How much interest is she charged? **1**

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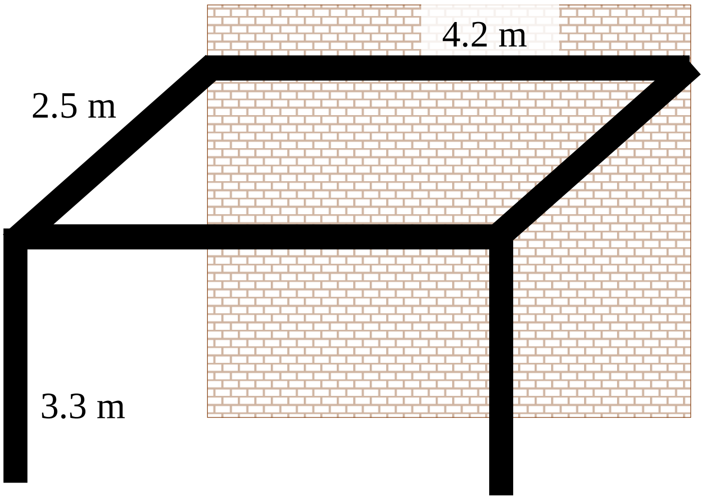
Question 15. (5 marks)

The plan for the timber beams of a pergola to be attached to an existing wall of a house are shown in the diagram.



(a) If timber costs $15 per linear metre. How much will the timber cost? **1**

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(b) The bill for the job is to be charged at cost of materials plus $60 per hour labour. How much does it cost if the job takes 2 hours? **1**

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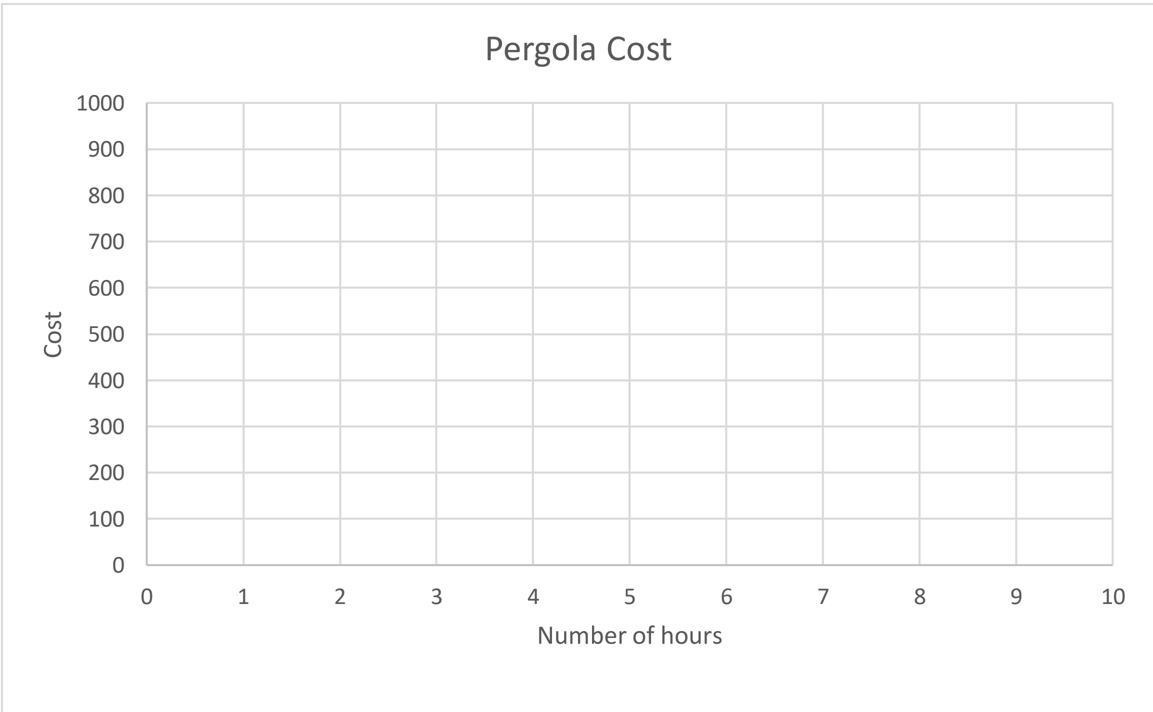
(c) Draw a graph to represent the bill for the job. **2**



(d) Estimate the cost of the job if it takes a total of 3.5 hours to complete. **1**

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Question 16. (3 marks)

The blood types of a 1000 people in a northern Australian town were recorded.

(a) Complete the table below **2**

Blood Type	Frequency	Relative Frequency
AB	30	0.03
A	380	
B	100	
O^-	90	
O^+	400	

(b) What is the probability that a person randomly selected from this town has a blood type O^-.

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Question 17. (2 marks)

Susie and Alex live in Sydney $(UTC+10)$ and just got engaged, Susie wants to phone her parents in New York $(UTC-5)$.

She wants to ring them between 6am and 8am on Monday morning local time in New York. Between what times should she ring from Sydney? **2**

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Question 18. (5 marks)

A survey was conducted of 50 people who were asked their average monthly mobile phone cost. The data is summarised in the following table.

Average Monthly Phone Cost	Class centre (x)	Frequency (f)	fx
1 - 50	25.5	6	153
51 - 100		7	
101 - 150		13	
151 - 200		11	
201 - 250		18	
251 - 300		5	
		$\sum f = 50$	$\sum fx =$

(a) Calculate the missing values in the class centre (x) and fx column. **2**

(b) Calculate $\sum fx$ and complete the table. **1**

(c) Calculate an approximation for the mean. **1**

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(d) Explain why the table only gives an approximation of the mean. **1**

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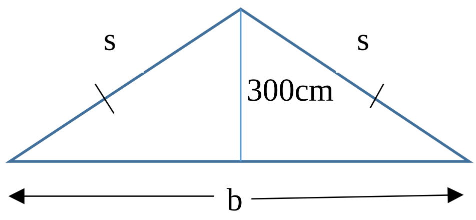
Question 19. (4 marks)

Jason measures the dimensions of the frame in the diagram below.



(a) For the base (b), Jason measures 656.2cm. Find the percentage error. **2**

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(b) If the height is to be extended to 400cm. Find the length of the two supporting beams (s) for the new frame, to 1 decimal place. **2**

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Question 20. (3 marks)

Henry bought a bag containing two types of lollies. The bag starts with fifty white lollies and seventy-five green lollies. He is only allowed to eat one lolly a day.

(a) What is the probability that the first lolly drawn is white? **1**

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Henry really wants a white lolly.

He draws two lollies at random with replacement.

(b) i) Show that the probability of drawing a green then a white is 0.24. **1**

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ii) If he repeats this experiment 25 times. How often can Harry expect to see a green lolly followed by a white lolly? **1**

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Question 21. (2 marks)

EP Electricity and Gas

Price summary

General charges

Daily supply charge: **71.50 cents**/day

General usage rates: **25.30 cents**/kWh

Solar feed-in

10.5 cents/kWh exported

Green power

10 to **100%** options available

Deanna has just installed solar panels on her roof. Her bill arrives and states that her General usage for 90 days is 2080 kWh. She has fed 390 kWh of energy back into the solar feed-in.

What is her bill for the 90 days? **2**

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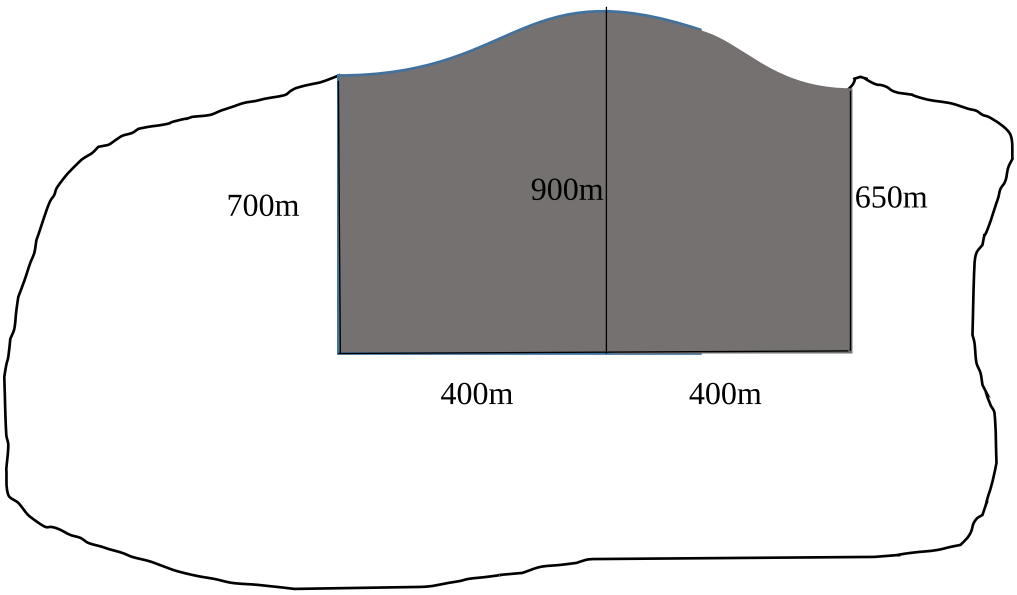
Question 22. (2 marks)

Green Island recently suffered extreme bushfires. Use two applications of the **Trapezoidal rule** to estimate the area of the island burnt out by bushfires shaded below. **2**



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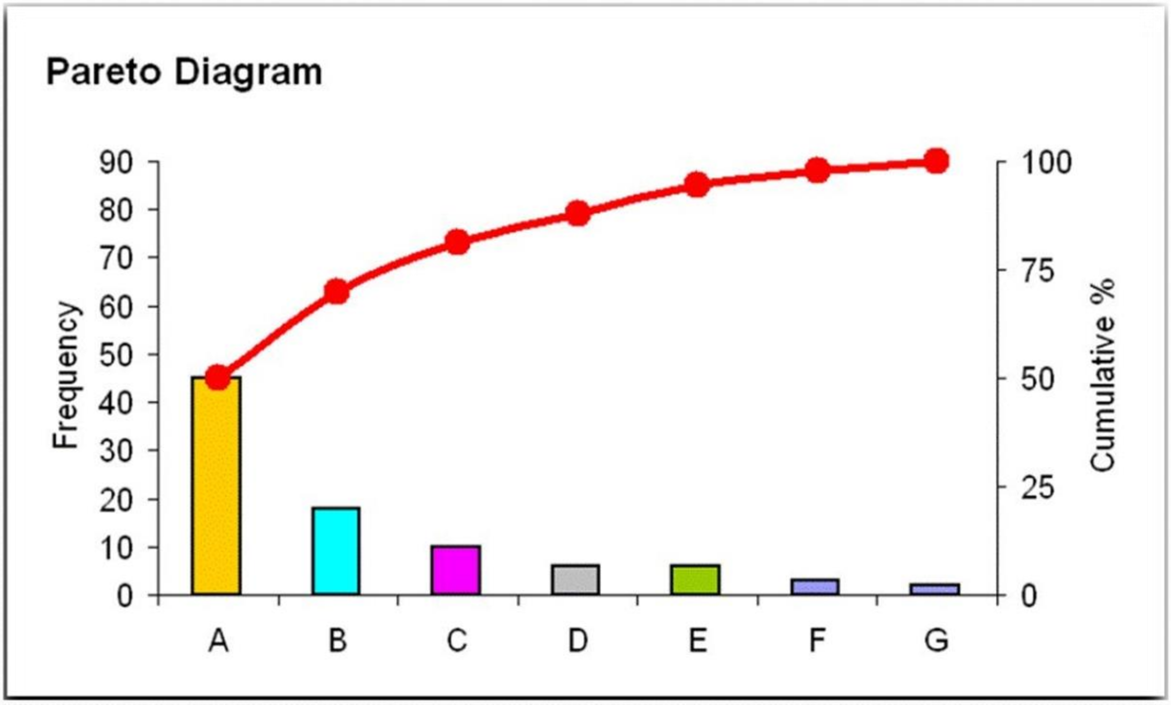
Question 23. (2 marks)



The graph above shows the results of a survey given to customers of a city hotel

Category A represents location, category B represents cleanliness, category C represents Service, category D represents size of room, E represents Noise, F represents access to parking and G represents other.

a) Location improved when a new tram stop was built near the hotel. The hotel decides to work on the category with the next highest number of complaints. If they improve this category approximately what percentage of customers would have their complaint addressed? **1**

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b) What additional category would they need to address to ensure they are satisfying the complaints of 80% of their customers? **1**

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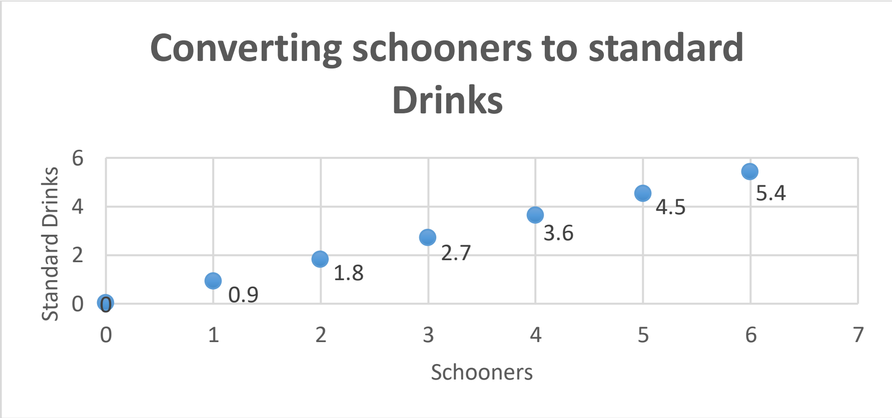
Question 24. (5 marks)

Azi goes to a party and consumes 4 schooners over 3 hours.



(a) If N is the number of standard drinks and H the number of hours, calculate, correct to two significant figures, Azi's blood alcohol content after 3 hours if his mass (M) is 81 kg. **2**

Use the formula,

$$BAC_{male} = \frac{10N - 7.5H}{6.8M}$$

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(b) Approximately how long, to the nearest minute, will it take for Azi's blood alcohol content to reach zero? **1**

Use the formula,

$$T = \frac{BAC}{0.015}$$

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(c) Using the graph, write a linear equation to represent the relationship between number of schooners (S) and number of standard drinks (N). **2**

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Question 25. (4 marks)

Minnie is employed in a café on a casual basis.

She earns \$22 per hour plus time-and-a-half on Saturday and double time on Sunday.

Here is her timesheet for last week.

Day	In	Out	Break	Number of Hours worked
Monday	Cafe closed			
Tuesday	9am	3pm	1 hour	5 hours
Wednesday	10am	2pm	Half hour	
Thursday	-	-	-	
Friday	7am	10am	No break	
Saturday	7:30am	2:30pm	Half hour	
Sunday	8:30am	2pm	Half hour	

(a) Complete the hours worked column. **2**

(b) Find how much she will be paid for the work? **2**

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Question 26. (9 marks)

Attendance at home games and away games for the 9 AFL teams in 2019 are listed in the table below (in 10 000's).

Home	48	27	51	64	52	45	36	12	13
Away	27	32	35	56	54	29	46	31	30

(a) The five-number summary for Home is shown in the table below

Complete the five-number summary for Away? **2**

	Minimum	Lower Quartile	Median	Upper Quartile	Maximum
Home	12	20	45	51.5	64
Away					

(b) Determine the range and interquartile range for each set of data **2**

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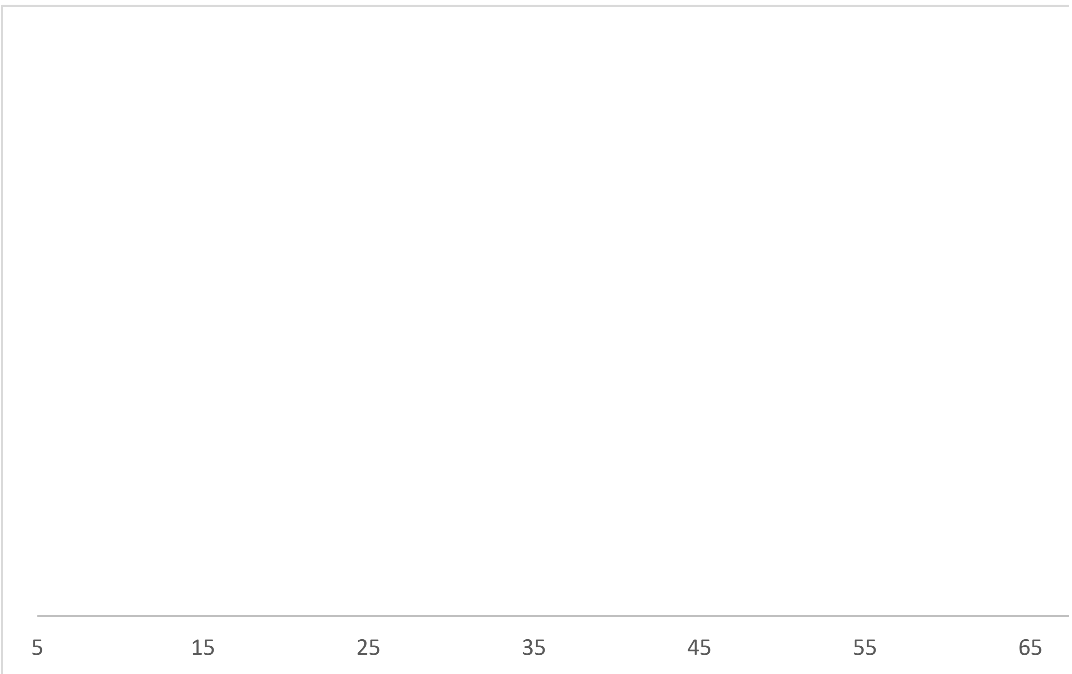
(c) Draw a parallel box plot for Home games and Away games, using the same number line. **3**



(d) Referring to spread and skewness, comment on any similarities or differences between the two sets of data. **2**

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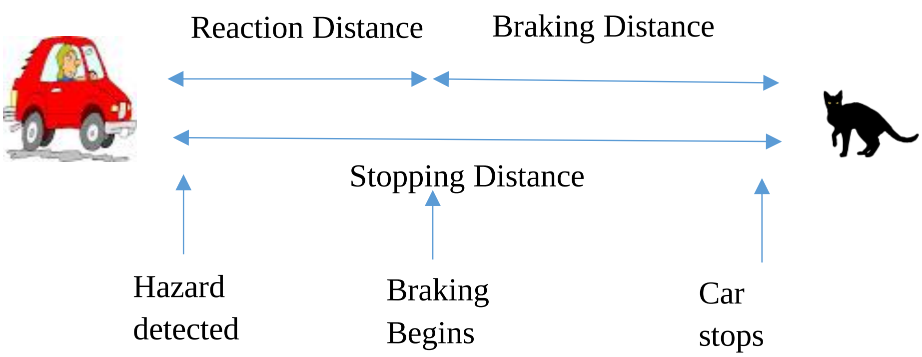
Question 27. (3 marks)

A cat ran out onto the road in front of Joni while she was driving home.

If Joni has travelled 30km in half an hour and her reaction time is 0.75.



(a) What is her average speed (in km/h)? **1**

(b) Use the stopping distance (d in metres) formula below to find Joni's stopping distance. **1**

$$d = \frac{5vt}{18} + \frac{v^2}{170}$$

where v is your speed in km/h and t is your reaction time.

(c) If the cat is 50m away, will she be able to stop in time? **1**

Question 28. (6 marks)

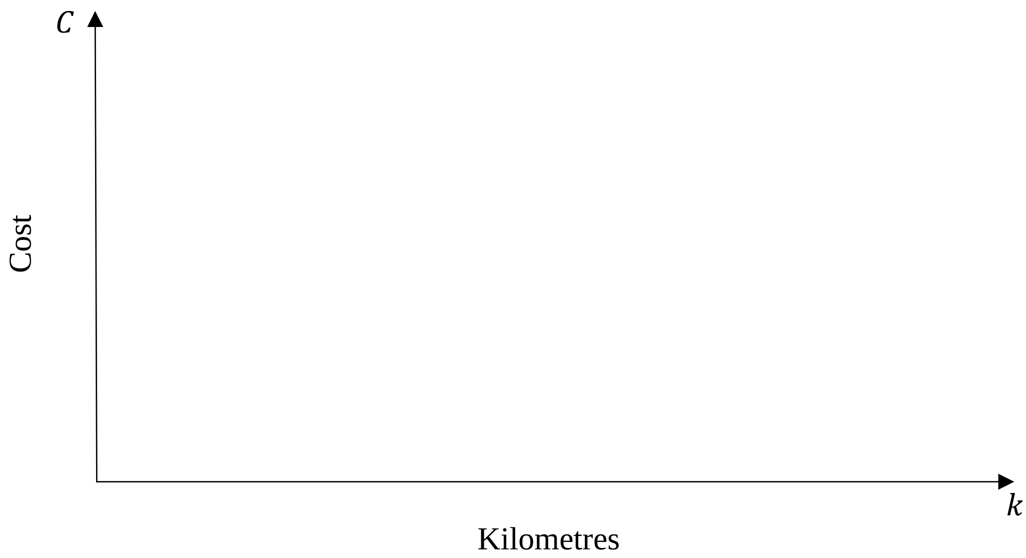
Fred wants to compare the cost of buying a car to taking an Uber to work each day. He works 5 days a week for 48 weeks of the year and it takes him 30 minutes each way to travel to work. Fred estimates the fixed costs for the car to be \$3744 per year including the loan, insurance, rego and maintenance plus weekly running costs of \$3 per kilometre.

(a) Write an equation to calculate the weekly cost (C) of the car in the form: **1**

$$\text{C} = \text{fixed cost } + \text{weekly running cost } \times \text{ number of kilometres } (k)$$

(b) If an Uber costs approximately \$6 per kilometre during the times when he is travelling to and from work write a linear equation to show the weekly cost of using an Uber. **1**

(c) Graph the two functions below and estimate the number of kilometres that Fred would have to travel for it to cost the same in his own car and an Uber. **3**



(d) If the taxation office allows Fred to claim 68 cents per kilometre for work related travel how much can he claim per year if the trip to work is 25km each way? **1**

Question 29. (2 marks)

a) The value of Vern's shares were valued at \$17506. They increased by 8%, then decreased by 3%, how much are they worth now? **1**

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b) How much profit, to the nearest dollar, did he make? **1**

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Question 30. (6 marks)

(a) Cindy is a single mother studying at university. She receives a \$450 fortnightly youth allowance payment and works part time earning \$823 a month. She has allowable deductions of \$200 a year. **2**

Calculate her taxable income.

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(b) Calculate Cindy's Medicare payment if the Medicare levy is 2% of your taxable income. **1**

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The table below is used to calculate tax payable.

Taxable income	Tax payable
\$0 – \$18 200	Nil
\$18 201 – \$37 000	19c for each \$1 over \$18 200
\$37 001 – \$87 000	\$3 572 plus 32.5c for each \$1 over \$37 000
\$87 001 – \$180 000	\$19 822 plus 37c for each \$1 over \$87 000
\$180 001 and over	\$54 232 plus 45c for each \$1 over \$180 000

(c) Cindy paid \$715 in PAYG tax during th financial year. Determine if she owes tax or will receive a tax refund, and by how much. **3**

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

REFERENCE SHEET

Measurement

Precision

Absolute error = $\frac{1}{2} \times$ precision

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}\left(x + y\right)$$

$$A \approx \frac{h}{2}\left(d_f + d_l\right)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$A = \frac{1}{2}ab\sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV\left(1 + r\right)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

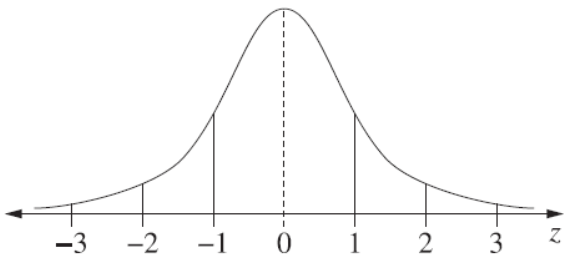
$$S = V_0\left(1 - r\right)^n$$

Statistical Analysis

$$z = \frac{x - \bar{x}}{s}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between –1 and 1
- approximately 95% of scores have z-scores between –2 and 2
- approximately 99.7% of scores have z-scores between –3 and 3

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MULTIPLE CHOICE ANSWER SHEET (12 marks)

Name: ……………………………………………….

Student Number: ……………………………………………….

Choose the best answer by filling in the circle corresponding to $A, B, C \text{or} D$.

	A	B	C	D
1	○	○	○	○
2	○	○	○	○
3	○	○	○	○
4	○	○	○	○
5	○	○	○	○
6	○	○	○	○
7	○	○	○	○
8	○	○	○	○
9	○	○	○	○
10	○	○	○	○
11	○	○	○	○
12	○	○	○	○